CHECKPOINT 1 · OCTOBER 20, 2026 · 15 MINUTES

Online Testing Application

Network Programming project · preliminary design and progress

Team: Trần Hữu Đức · Trần Anh Dũng · Trần Cao Phong · Lê Diễm Hằng

Objectives and main functions

Test Mode

- Create room: question count and duration.
- View rooms; join before start.
- Owner starts/ends; users can revise answers.
- Submit or timeout → score and saved results.

Practice and shared features

- Choose count/time; practice and view score.
- Register/login and room permissions.
- Filter questions by topic and difficulty.
- GUI, attempt history, and score charts.

Tools and system architecture



> **Tools:** GCC, pthreads, cJSON, SQLite, libsodium, Make, Docker Compose.

TCP + JSON Lines between client and server. The server manages sessions, rooms, deadlines, and scoring.

Application-layer protocol

TCP + UTF-8 JSON Lines. \n denotes the LF byte 0x0A.

Client → Server

```
{"type":"ATTEMPT_SUBMIT","request_id":42,"data":{"attempt_id":11}}\n
```

Server → Client (event)

```
{"type":"EVENT","event":"TEST_FINISHED","data":{"test_id":41,"reason":"TIMEOUT"}}\n
```

> Buffer reads until LF; use `send_all()` for partial writes. Only the server validates and scores answers.

Commands and access rules

Accounts and rooms

`REGISTER · LOGIN`

`ROOM_CREATE · ROOM_LIST`

`ROOM_JOIN · ROOM_START`

`ROOM_END`

Login binds identity to the connection. Only the owner starts/ends a room.

Attempts and results

`QUESTIONS_GET · ANSWER_SET`

`ATTEMPT_SUBMIT`

`ROOM_RESULTS_GET`

`HISTORY_GET · STATS_GET`

Only your own attempt accepts answers. Responses carry data or a stable error code.

Test and practice workflows

Test room lifecycle

NOT STARTED
Create · join

→

ONGOING
Answer · revise · submit

→

FINISHED
Score · results

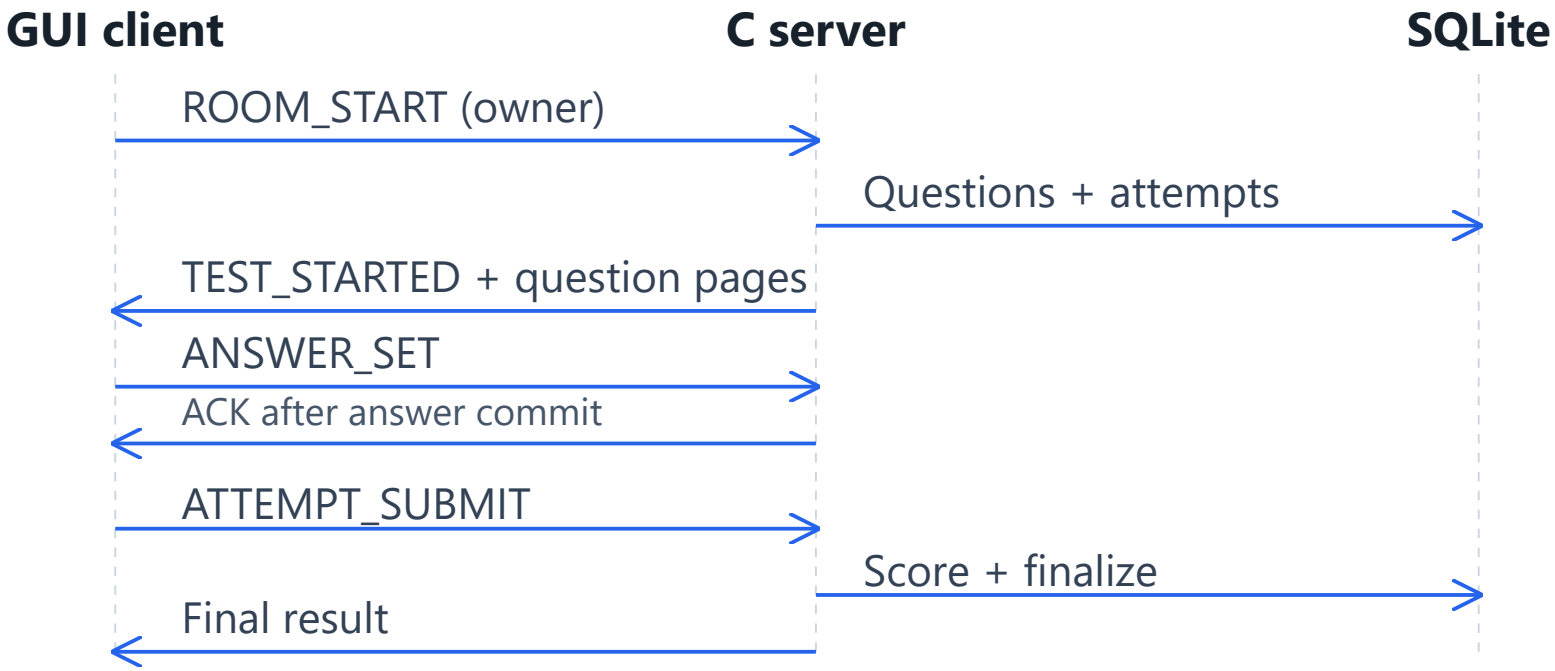
Test Mode

Owner controls start/end. Join before start. Server enforces the deadline.

Practice Mode

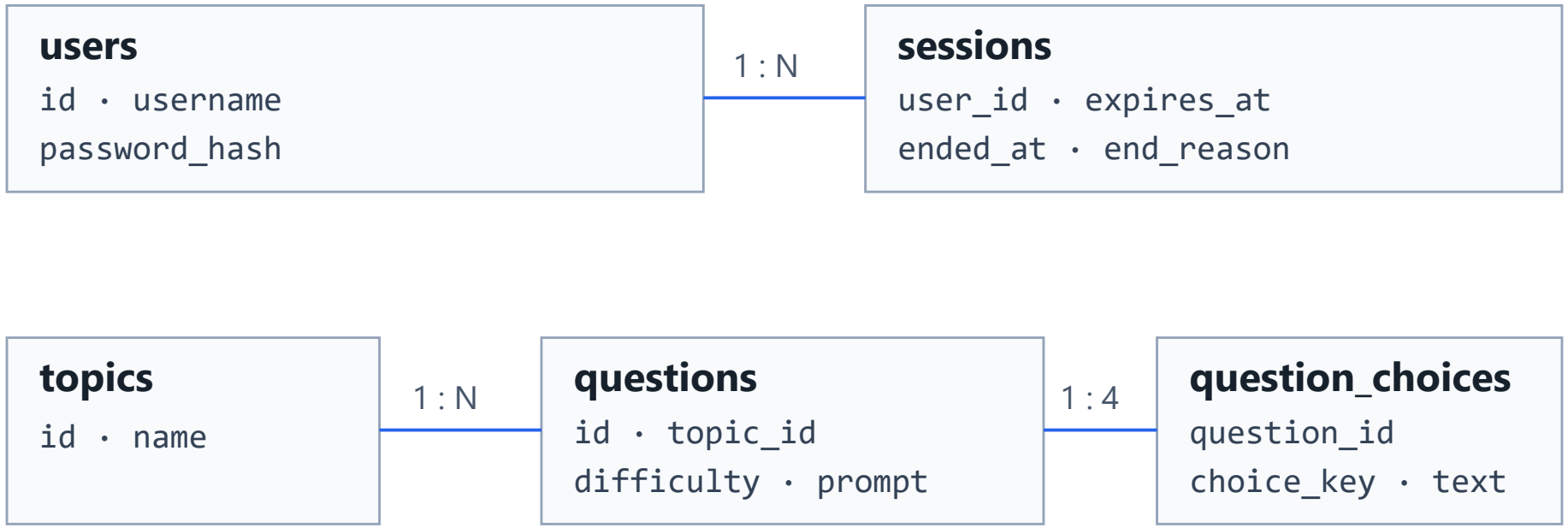
Choose count/time → answer → submit or timeout → score.

Test message sequence



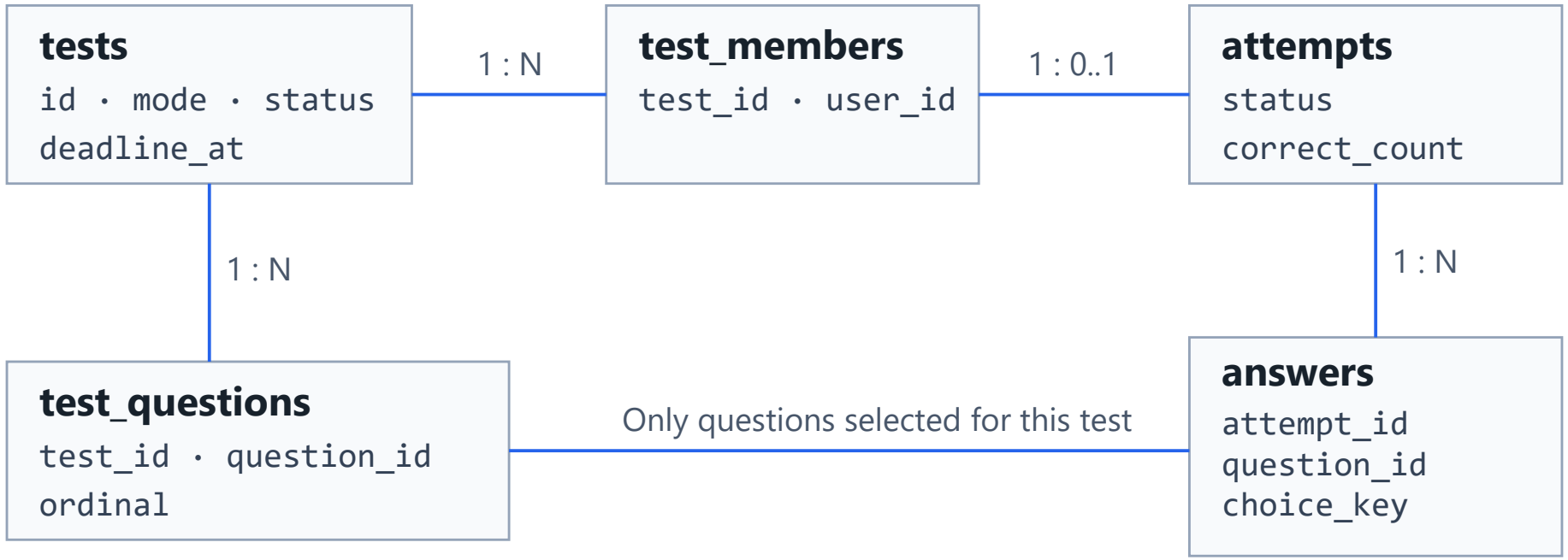
Timeout and owner-end use the same finalization path. Saved answers are committed before acknowledgement.

Accounts and question bank



Four choices, one correct key. Used questions stay immutable so stored results remain valid.

Shared test, individual attempts



Room = many attempts. Practice = one attempt. Topic/difficulty filter tables and activity logs complete the schema.

Question filters and statistics

Question bank

Filter by topic and difficulty; choose question count.

Store: questions, attempts, answers, scores, and activity logs.

The chart uses sample values only; no real user results exist yet.



Task allocation and current status

Member	Responsibility
Trần Hữu Đức · 20225963	C TCP sockets and room lifecycle
Trần Anh Dũng · 20226031 · Project lead	Coordinate architecture, protocol, integration, and demo
Trần Cao Phong · 20226061	Accounts, access control, SQLite, activity log
Lê Diễm Hằng · 20225970	Tkinter client, history/statistics, end-to-end tests

Designed / checked

Flows, ERD, SQL schema, protocol; 12 schema checks

Next

C TCP framing → accounts → attempts and rooms

Not implemented

C application handlers and GUI